



DEPARTMENT OF SOFTWARE ENGINEERING

Cloud Computing Lab 01

Submitted By: **Mahnoor Khan**

Semester :5th Semester

Registration No:**Fa24b1-Se-012**

Group A

Course Code :**Cloud Computing (SE-665)**

Github Username :**mahnoorkhan5376**

Task 1: Create and Configure Your GitHub Account

**Mahnoor Khan** (mahnoorkhan5376)
Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Change username

Changing your username can have [unintended side effects](#).

Change username

? Looking to manage account security settings? You can find them in the [Password and authentication](#) page.

Link Patreon account

Connect a Patreon account for @mahnoorkhan5376 to sponsor maintainers with. Get recognition on GitHub for sponsorships made on Patreon when the sponsored person has linked Patreon and GitHub, too, and has a public GitHub Sponsors profile.

Connect with Patreon

GitHub Account settings confirming username: mahnoorkhan5376

**Mahnoor Khan** (mahnoorkhan5376)
Your personal account

Public profile

Account

Appearance

Accessibility

Notifications

Access

Billing and licensing

Emails

Password and authentication

Sessions

SSH and GPG keys

Credentials

Organizations

Public profile

Name

Mahnoor Khan

Your name may appear around GitHub where you contribute or are mentioned. You can remove it at any time.

Public email

Select a verified email to display

You can manage verified email addresses in your [email settings](#)

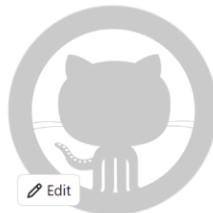
Bio

Tell us a little bit about yourself

You can @mention other users and organizations to link to them.

Pronouns

Don't specify

Profile picture

Edit

Task 2: Register in the Student Marks Portal



Student Marks Portal

Sign up or sign in securely with your GitHub account.

Continue with GitHub

Waiting for GitHub sign-in...

Student Marks Portal login page with "Continue with GitHub" sign-in



Mahnoor Khan

GitHub authenticated

Registration
2024-BSE-12

GitHub username
[mahnoorkhan5376](#)

Email
kashmala5376@gmail.com

Approved courses
1

+ Enroll in another course

APPROVED COURSES

Course Marks

Not published

Cloud Computing (SE-665) Section A

ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
------------	----------	-------	------------

Marks have not been published.
Portal profile showing GitHub-authenticated account, registration number, and enrolled course

+ Enroll in another course

APPROVED COURSES

Course Marks

Not published

Cloud Computing (SE-665) Section A

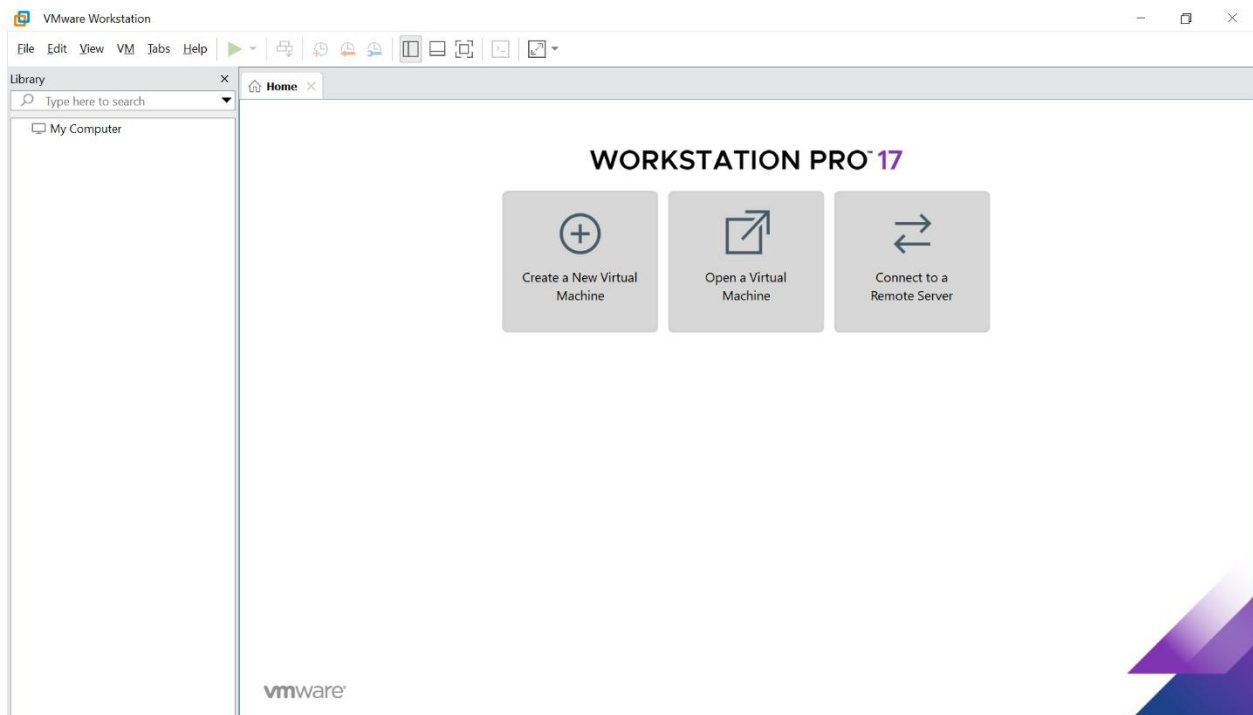
ASSESSMENT	OBTAINED	TOTAL	PERCENTAGE
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Marks have not been published.

Total

Student Marks Portal enrollment summary with total marks row

Task 3: Download VMware Workstation Pro and Ubuntu Server



VMware Workstation Pro running successfully on Windows

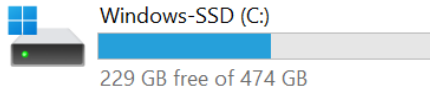
▼ Today

 ubuntu-26.04.1-live-server-amd64 (1)	9/14/2026 9:24 PM	Disc Image File	2,859,240 KB
 ubuntu-26.04.1-live-server-amd64	9/14/2026 9:22 PM	Disc Image File	2,859,240 KB

▼ Last week

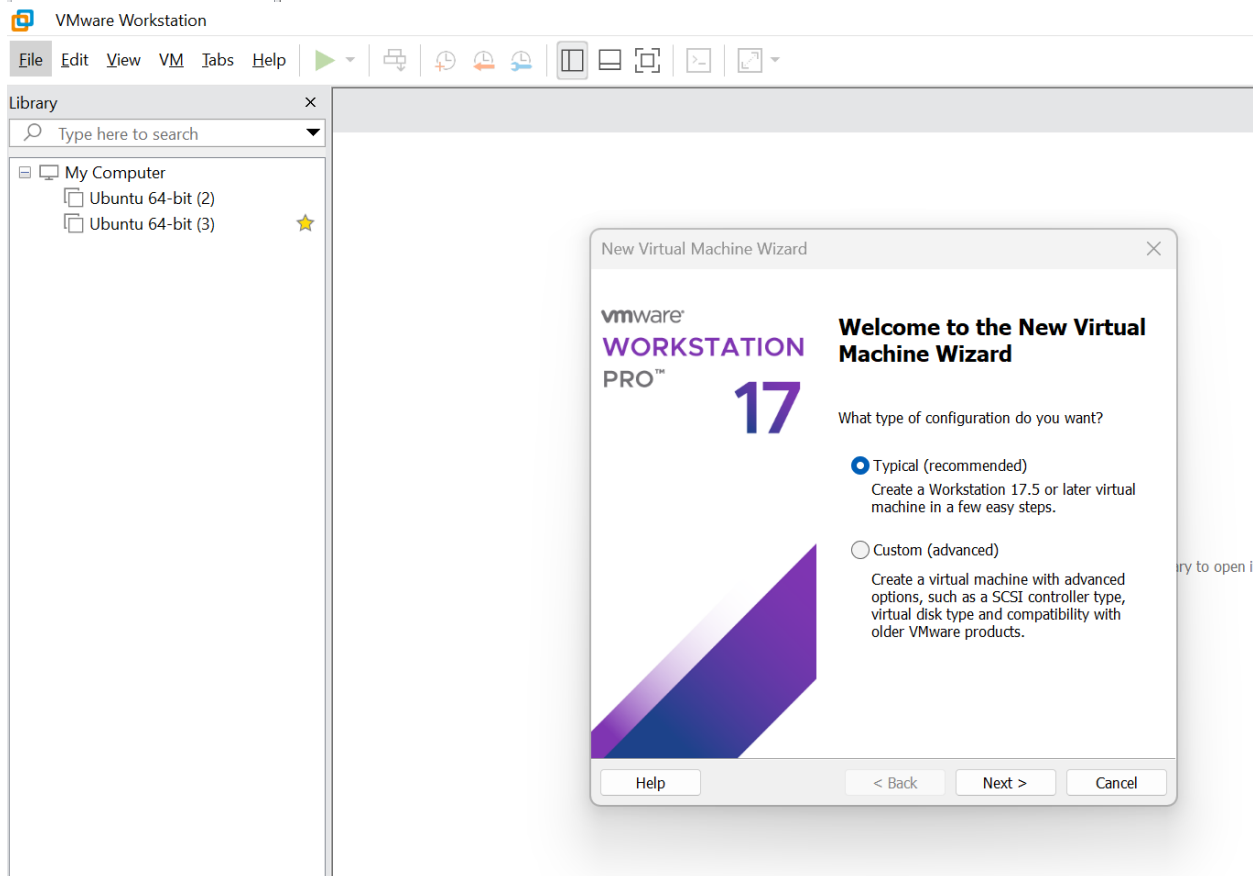
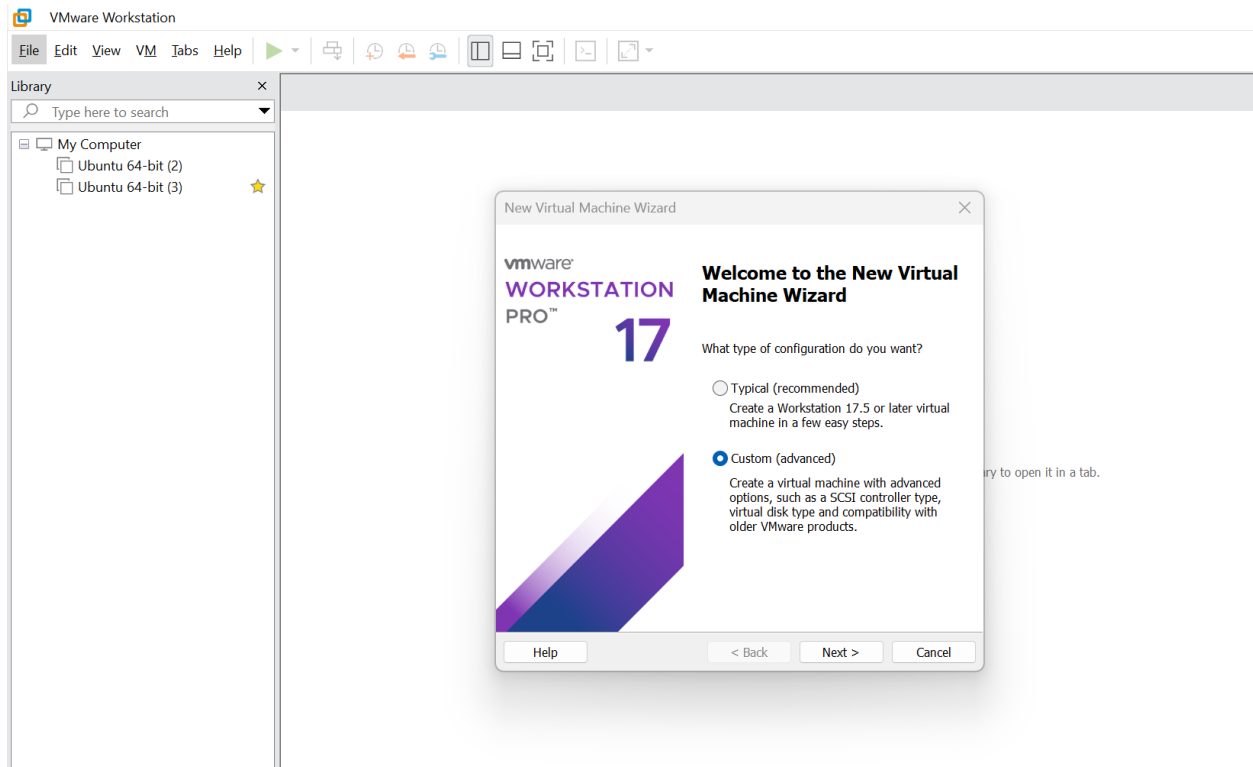
Downloaded Ubuntu Server LTS ISO file shown in the Downloads folder

▼ Devices and drives



Available disk storage confirming sufficient free space for the VM

Task 4: Create the Ubuntu Server Virtual Machine



New Virtual Machine Wizard: Custom vs. Typical (recommended) configuration, with Typical selected

New Virtual Machine Wizard

Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:
No drives available

☒ Installer disc image file (iso):
C:\Users\User\Downloads\ubuntu-26.04.1-live-server-a Browse...

 Ubuntu 64-bit Server 26.04.1 detected.
To use Easy Install, insert the first disc of the set.

☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel

Ubuntu Server ISO file path selected in the VM wizard

WORKSTATION PRO™ 17

New Virtual Machine Wizard

Ready to Create Virtual Machine
Click Finish to create the virtual machine and start installing Ubuntu 64-bit.

The virtual machine will be created with the following settings:

Name:	Ubuntu 64-bit (5)
Location:	C:\Users\User\Documents\Virtual Machines\Ubuntu 64-bit ...
Version:	Workstation 17.5 or later
Operating System:	Ubuntu 64-bit
Hard Disk:	20 GB, Split
Memory:	2048 MB
Network Adapter:	NAT
Other Devices:	2 CPU cores, CD/DVD, USB Controller, Sound Card

Customize Hardware...

☒ Power on this virtual machine after creation

< Back Finish Cancel



Connect to a
Remote Server

Final virtual machine configuration summary (20 GB disk, NAT network)

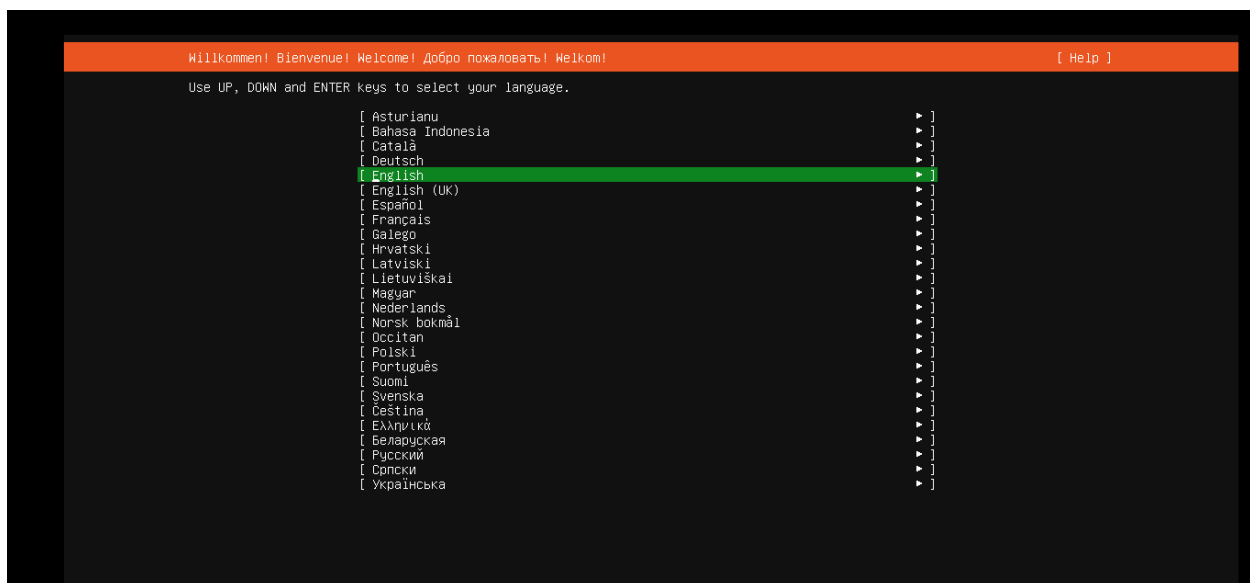
```

Starting dbus.service - D-Bus System Message Bus...
[ OK ] Started dmesg.service - Save initial kernel messages after boot.
Starting dracut-shutdown.service - Restore /run/initramfs on shutdown...
Starting e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots...
[ OK ] Reached target getty.target - Login Prompts.
Starting networkd-dispatcher.service - Dispatcher daemon for systemd-networkd...
Starting nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot...
Starting polkit.service - Authorization Manager...
Starting rsyslog.service - System Logging Service...
[ OK ] Reached target getty-pre.target - Preparation for Logins.
Starting snapd.seeded.service - Wait until snapd is fully seeded...
Starting snapd.service - Snap Daemon...
Starting ssh.service - OpenBSD Secure Shell server...
Starting sysstat.service - Resets System Activity Logs...
Starting systemd-logind.service - User Login Management...
Starting udisks2.service - Disk Manager...
[ OK ] Finished blk-availability.service - Availability of block devices.
[ OK ] Finished dracut-shutdown.service - Restore /run/initramfs on shutdown.
[ OK ] Stopped ssh.service - OpenBSD Secure Shell server.
Starting ssh.service - OpenBSD Secure Shell server...
[ OK ] Finished nvme-autoconnect.service - Connect NVMe-oF subsystems automatically during boot.
[ OK ] Reached target remote-fs-pre.target - Preparation for Remote File Systems.
[ OK ] Reached target remote-fs.target - Remote File Systems.
Starting apport.service - automatic crash report generation...
[ OK ] Started cron.service - Regular background program processing daemon.
Starting multipathd-queueing.service - Enable queuing for multipath maps...
Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Finished multipathd-queueing.service - Enable queuing for multipath maps.
[ OK ] Finished sysstat.service - Resets System Activity Logs.
[ OK ] Started ssh.service - OpenBSD Secure Shell server.
[ OK ] Finished cloud-config.service - Cloud-init: Config Stage.
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
Starting plymouth-quit-wait.service - Hold until boot process finishes up...
Starting plymouth-quit.service - Terminate Plymouth Boot Screen...
[ OK ] Finished plymouth-quit-wait.service - Hold until boot process finishes up.
Starting setvtrgb.service - Set console scheme...
[ OK ] Finished plymouth-quit.service - Terminate Plymouth Boot Screen.
[ OK ] Finished e2scrub_reap.service - Remove Stale Online ext4 Metadata Check Snapshots.
[ OK ] Started systemd-logind.service - User Login Management.
[ OK ] Started unattended-upgrades.service - Unattended Upgrades Shutdown.
[ OK ] Finished setvtrgb.service - Set console scheme.
[ OK ] Started rsyslog.service - System Logging Service.
[ OK ] Started chrony.service - chrony, an NTP client/server.
[ OK ] Started dbus.service - D-Bus System Message Bus.
[ OK ] Started polkit.service - Authorization Manager.
Starting ModemManager.service - Modem Manager...
[ OK ] Started udisks2.service - Disk Manager...

```

Ubuntu Server installer boot screen output

Task 5: Choose the Installation Language



Installation language selection screen with English highlighted

Task 6: Configure the Keyboard Layout

Keyboard configuration

Please select your keyboard layout below, or select "Identify keyboard" to detect your layout automatically.

Layout:

Variant:

(+) English (UK)

() English (UK) - English (UK, Colemak)

() English (UK) - English (UK, Dvorak)

() English (UK) - English (UK, Dvorak, with UK punctuation)

() English (UK) - English (UK, Macintosh)

() English (UK) - English (UK, extended, with Win keys)

() English (UK) - English (UK, intl., Macintosh)

() English (UK) - English (UK, intl., with dead keys)

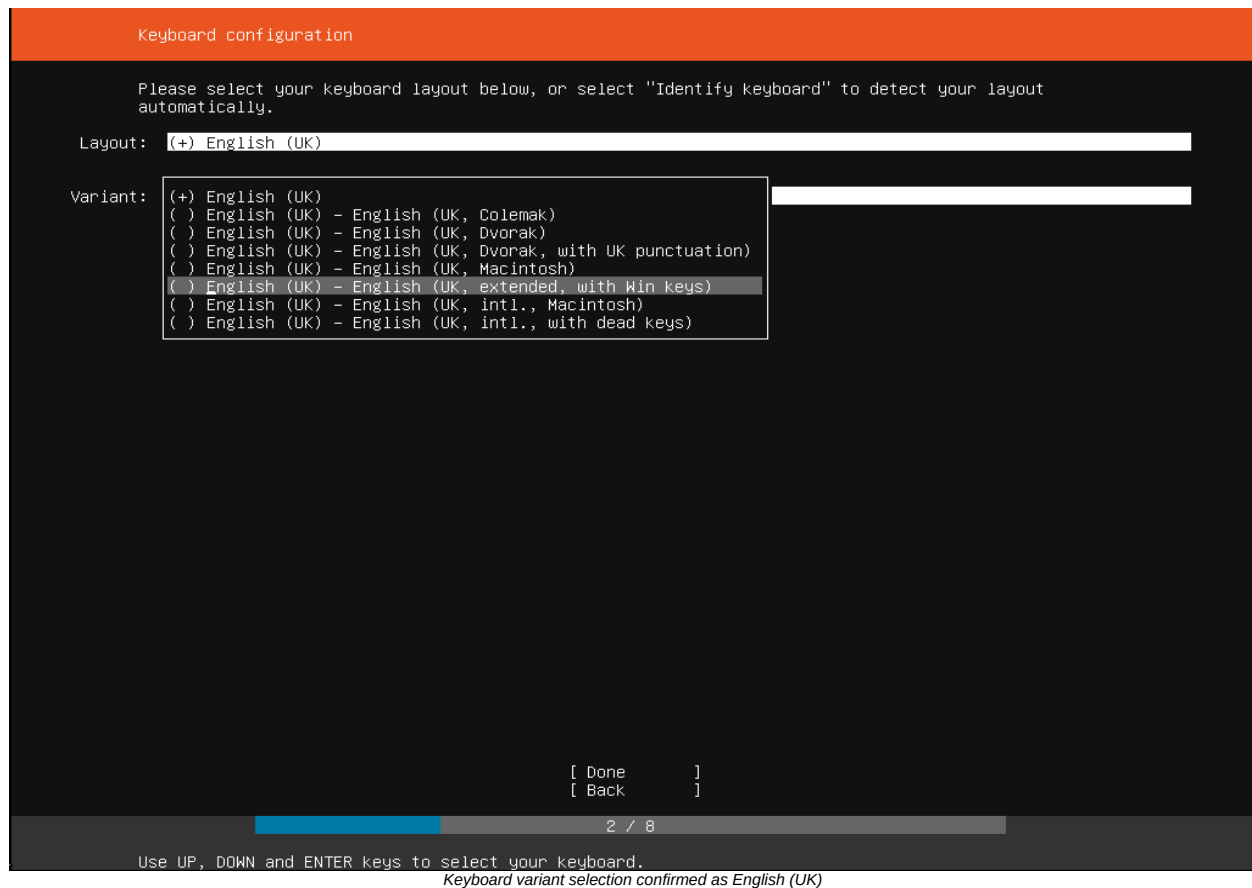
[Done]

[Back]

2 / 8

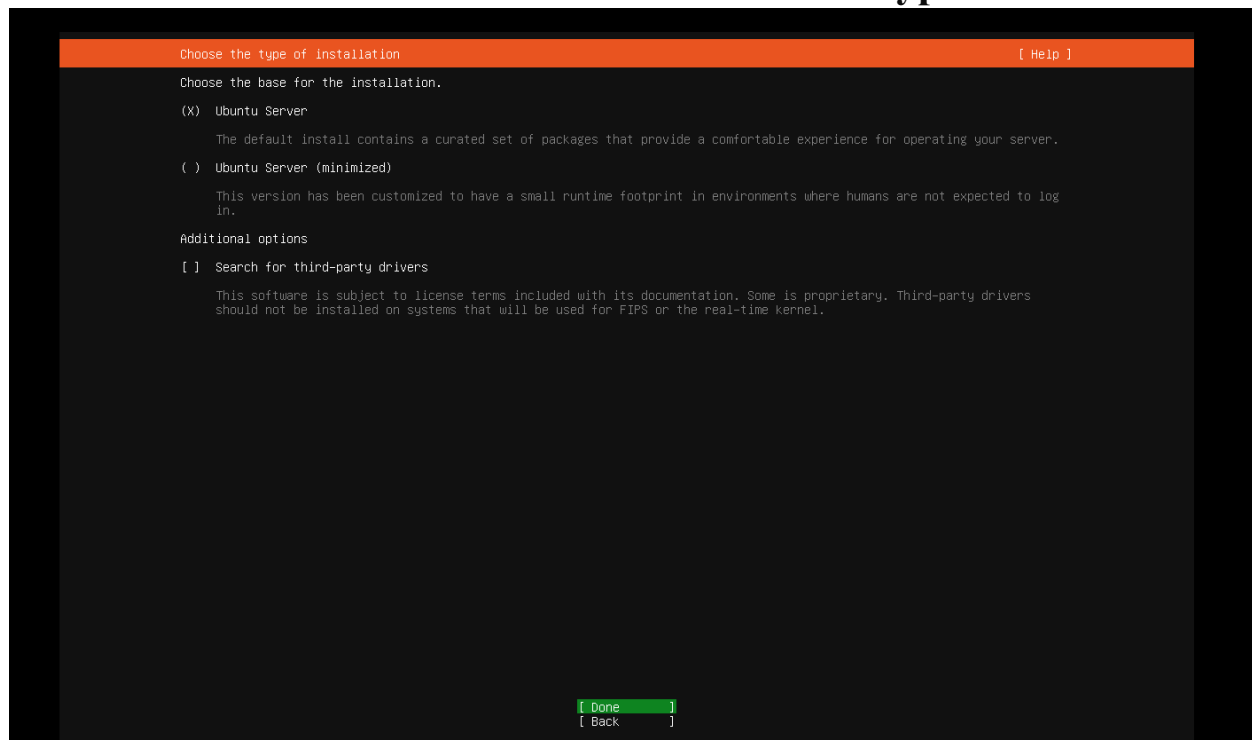
Use UP, DOWN and ENTER keys to select your keyboard.

Keyboard layout and variant configured as English (UK)



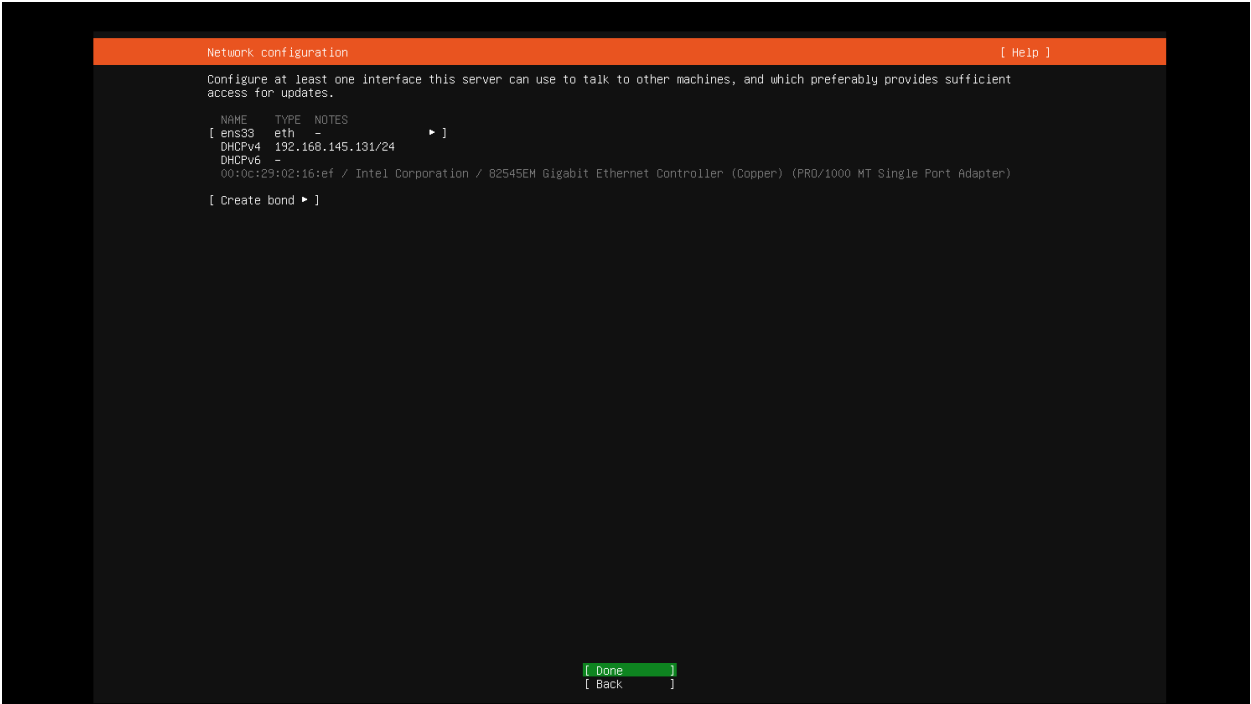
Keyboard variant selection confirmed as English (UK)

Task 7: Choose the Ubuntu Server Installation Type

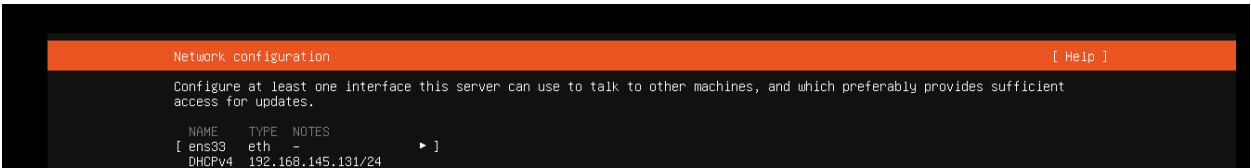


Ubuntu Server installation type selected, Done highlighted

Task 8: Configure the Network



Detected network interface (ens33) with DHCPv4 configuration



Network interface assigned an IP address automatically via DHCP

Task 9: Configure Storage

Guided storage configuration

[Help]

Configure a guided storage layout, or create a custom one:

☒ Use an entire disk

[/dev/sda local disk 20.00GB ▼]

☒ Set up this disk as an LVM group

☐ Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

☐ Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

☐ Custom storage layout

[Done]
[Back]

Guided storage configuration selecting the VMware virtual disk

Guided storage configuration

[Help]

Configure a guided storage layout, or create a custom one:

☒ Use an entire disk

[/dev/sda local disk 20.00GB ▼]

☒ Set up this disk as an LVM group

☐ Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

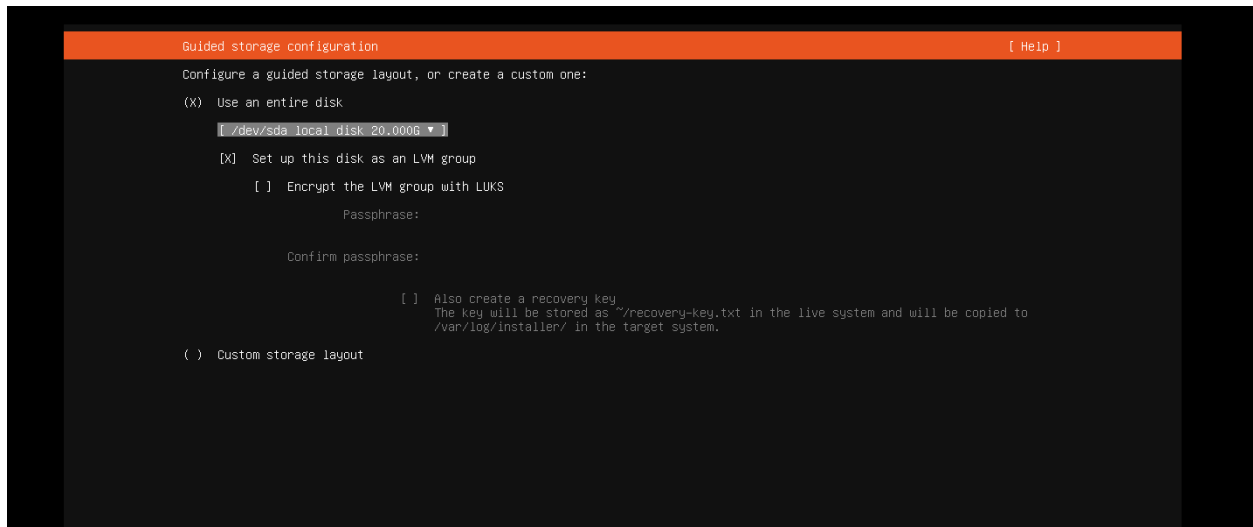
☐ Also create a recovery key
The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

☐ Custom storage layout

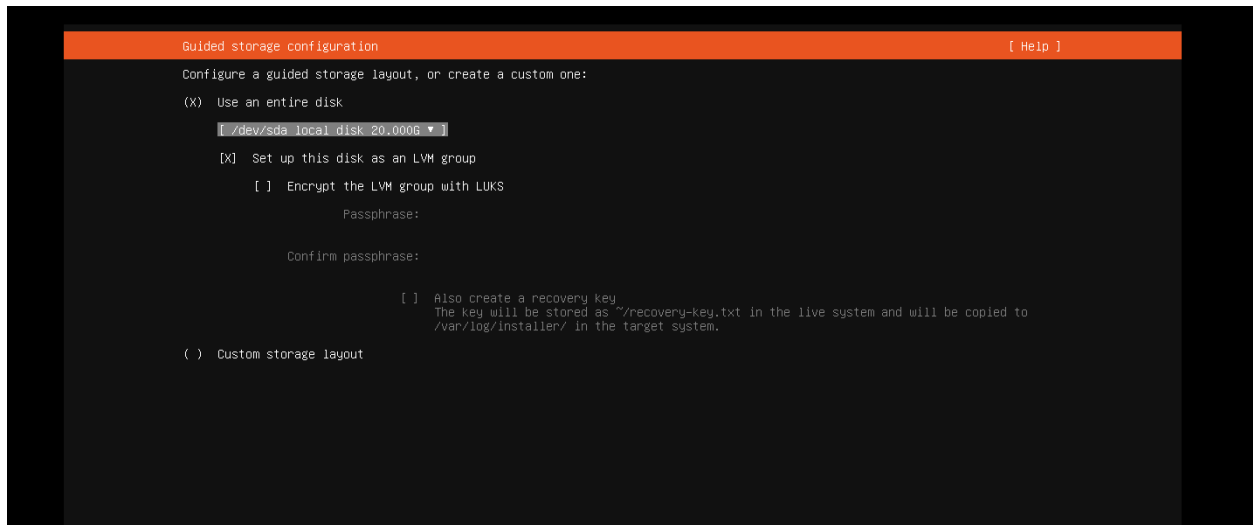
[Done]
[Back]

Storage configuration reviewed before continuing

Task 10: Select the Installation Disk

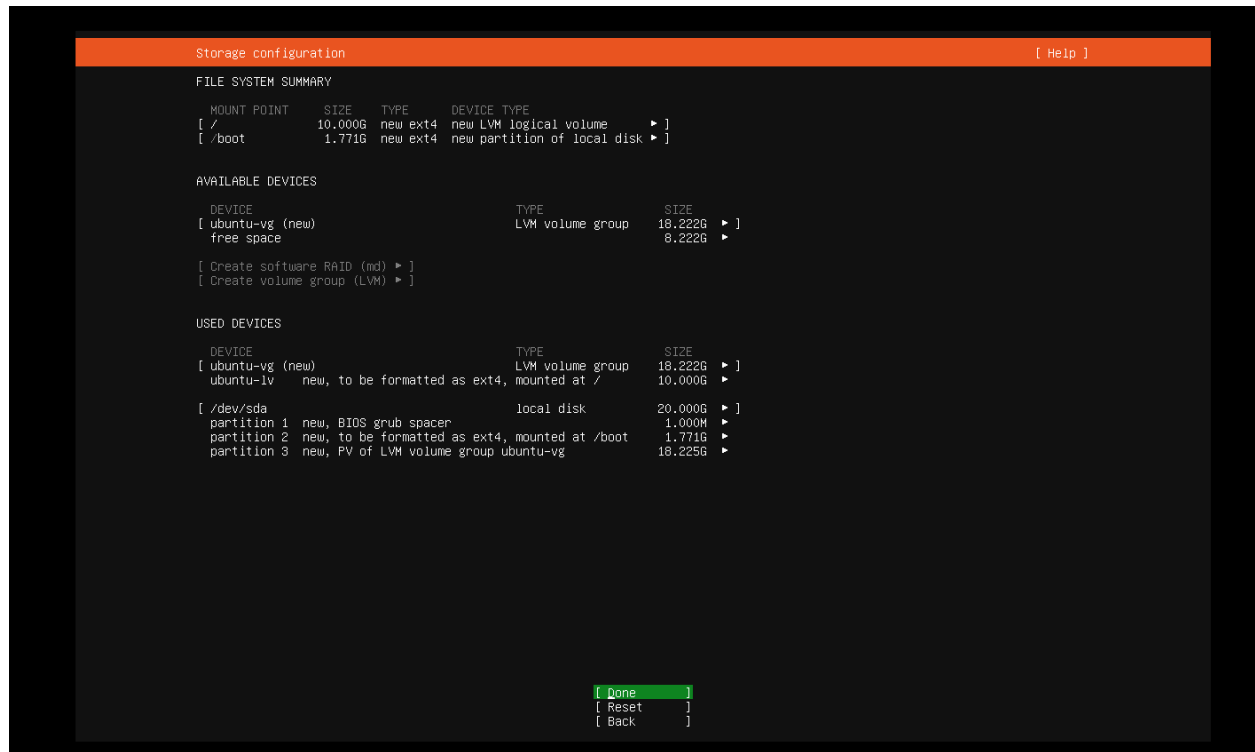


Selected installation disk: /dev/sda, 20,000G VMware virtual disk



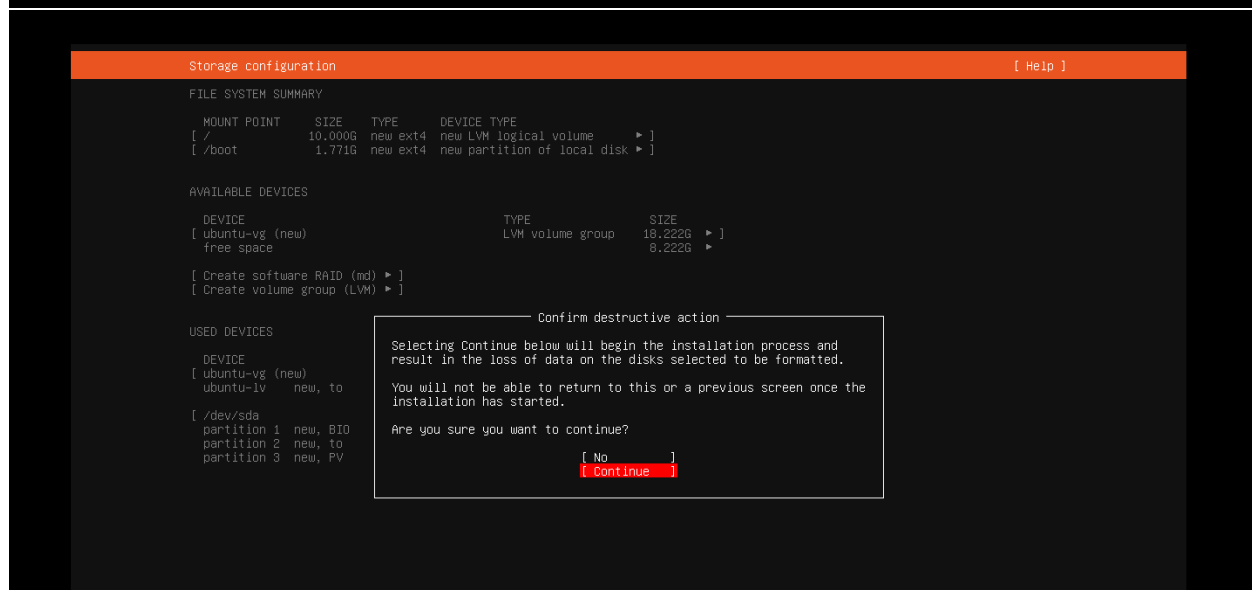
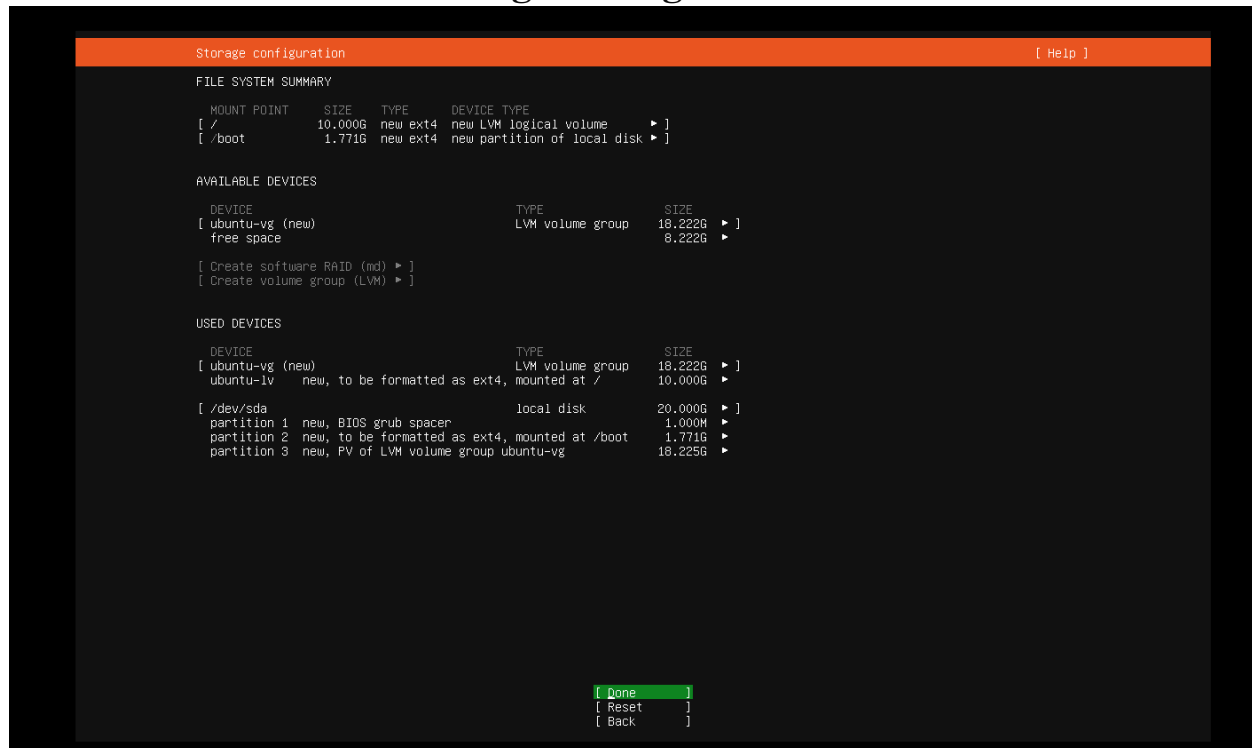
Disk name and size confirmed before proceeding

Task 11: Review the Partitions



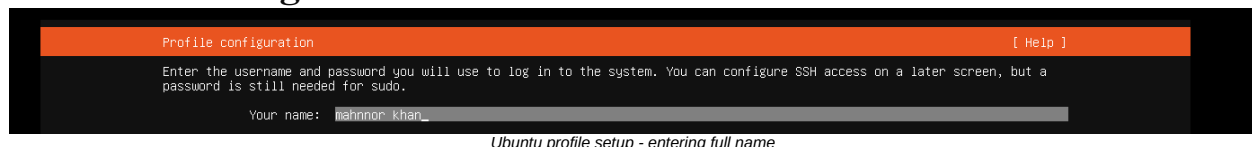
Proposed partition layout for the VMware virtual disk (/boot, LVM)

Task 12: Confirm the Storage Changes



Storage configuration reviewed, then Confirm destructive action dialog with Continue selected

Task 13: Configure the Ubuntu Profile



Ubuntu profile setup - entering full name

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: mahnor khan

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: mahnoorkhan5376_

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: mahnor khan

Your servers name: ubuntu
The name it uses when it talks to other computers.

Server name and username fields set (ubuntu / mahnoorkhan5376)

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name: mahnor khan

Your servers name: ubuntu
The name it uses when it talks to other computers.

Pick a username: mahnoorkhan5376

Choose a password: *****

Confirm your password: *****

[Done]

Completed Ubuntu profile with name, server name, username, and password (masked)

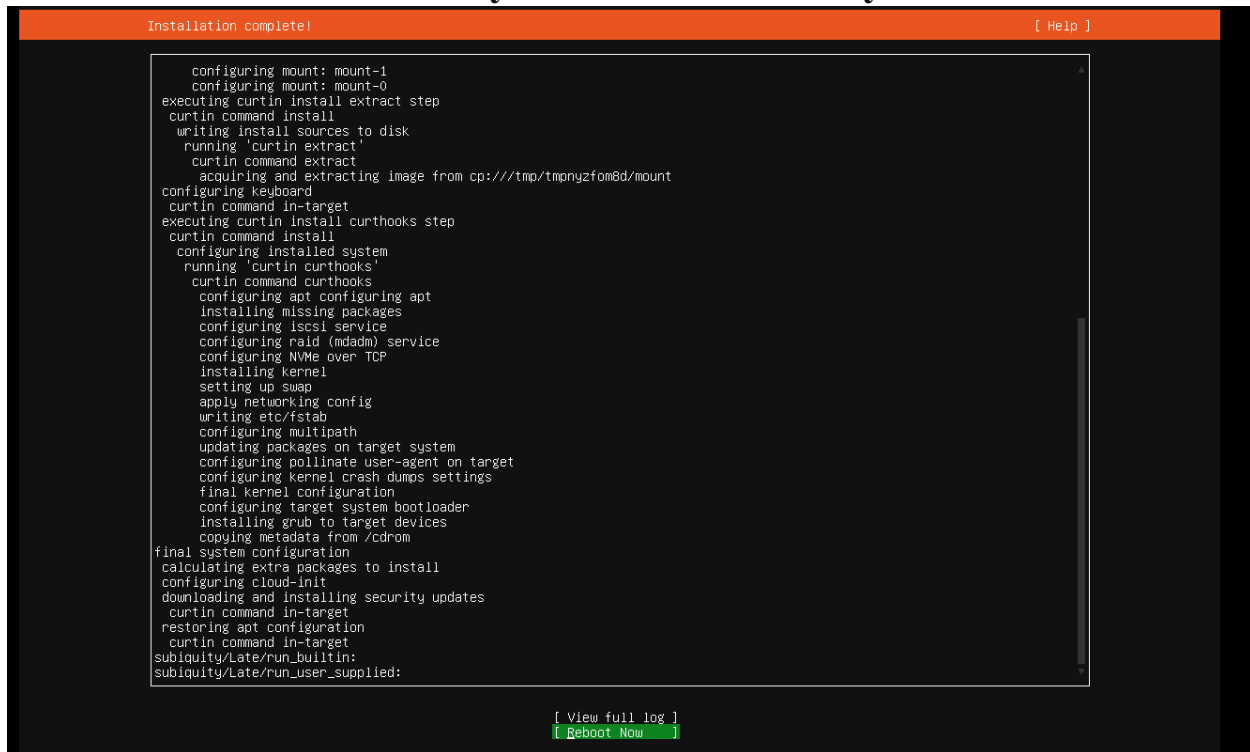
Task 14: Complete the Software Installation

```
configuring partition: partition-2
configuring format: format-0
configuring lvm_voigroup: lvm_voigroup-0
configuring lvm_partition: lvm_partition-0
configuring format: format-1
configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpnyzfom8d/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
```

[View full log]

Ubuntu Server software installation progress log

Task 15: Reboot and Verify the Ubuntu Identity

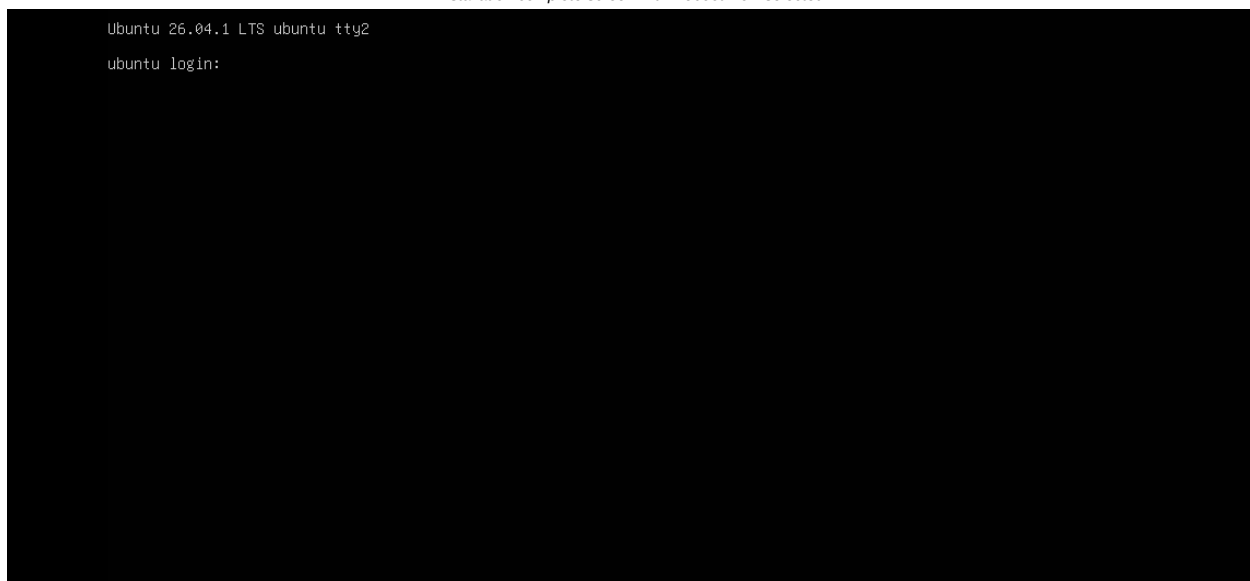


```
Installation complete! [ Help ]

configuring mount: mount-1
configuring mount: mount-0
executing curtin install extract step
curtin command install
writing install sources to disk
running 'curtin extract'
curtin command extract
acquiring and extracting image from cp:///tmp/tmpnyzfom8d/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
configuring installed system
running 'curtin curthooks'
curtin command curthooks
configuring apt configuring apt
installing missing packages
configuring iscsi service
configuring raid (mdadm) service
configuring NVMe over TCP
installing kernel
setting up swap
apply networking config
writing etc/fstab
configuring multipath
updating packages on target system
configuring pollinate user-agent on target
configuring kernel crash dumps settings
final kernel configuration
configuring target system bootloader
installing grub to target devices
copying metadata from /cdrom
final system configuration
calculating extra packages to install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
curtin command in-target
subiquity/late/run_builtins:
subiquity/late/run_user_supplied:
```

[View full log]
[Reboot Now]

Installation complete screen with Reboot Now selected



```
Ubuntu 26.04.1 LTS ubuntu tty2

ubuntu login:
```

Ubuntu Server login screen after reboot

```

Ubuntu 26.04.1 LTS ubuntu tty2

ubuntu login: mahnoorkhan5376
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Sep 15 06:19:48 PM UTC 2026

System load:  0.0               Processes:            252
Usage of /:   51.7% of 9.75GB   Users logged in:     0
Memory usage: 23%              IPv4 address for ens33: 192.168.145.132
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

8 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

mahnoorkhan5376@ubuntu:~$

```

Successful terminal login showing prompt mahnoorkhan5376@ubuntu:~\$

```

mahnoorkhan5376@ubuntu:~$

```

Ubuntu identity verified: prompt follows <github-username>@ubuntu format

Task 16: Find the Ubuntu Server IP Address

```

mahnoorkhan5376@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:02:16:ef brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c290216ef
    inet 192.168.145.132/24 metric 100 brd 192.168.145.255 scope global dynamic ens33
        valid_lft 1096sec preferred_lft 1096sec
    inet6 fe80::20c:29ff:fe02:16ef/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
mahnoorkhan5376@ubuntu:~$ _

```

"ip addr" command output showing the assigned IPv4 address

```
mahnoorkhan5376@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:02:16:ef brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c290216ef
    inet 192.168.145.132/24 metric 100 brd 192.168.145.255 scope global dynamic ens33
        valid_lft 1096sec preferred_lft 1096sec
    inet6 fe80::20c:29ff:fe02:16ef/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
mahnoorkhan5376@ubuntu:~$
```

IPv4 address 192.168.145.132 identified for the ens33 interface

Task 17: Connect to Ubuntu Server from Windows

```
C:\Users\User>ssh mahnoorkhan5376@192.168.145.132
```

SSH command run from Windows Command Prompt

```
C:\Users\User>ssh mahnoorkhan5376@192.168.145.132
The authenticity of host '192.168.145.132 (192.168.145.132)' can't be established.
ED25519 key fingerprint is SHA256:TZgSl4ZCHSerIsYSzON7tLhoM8dS56XY1lLSCESHsOg.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.145.132' (ED25519) to the list of known hosts.
```

SSH host fingerprint prompt accepted

```
mahnoorkhan5376@ubuntu:~$
```

Remote SSH session showing prompt mahnoorkhan5376@ubuntu:~\$